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DEPARTMENT OF HEALTH PAPUA NEW GUINEA

CONTRACTOR'S ENVIRONMENTAL MANAGEMENT PLAN

PROJECT:	Construction and Completion of Agavairu Health Centre Stages I and II in Kairuku Hiri District, Central Province.			
EMPLOYER:	National Department of Health			
PROJECT:	Health Services Sector Development Project			
COUNTRY:	Papua New Guinea			
PREPARED BY:	Jeremiah Rockey – Estimator/Senior Projects Engineer			
DUE DATE:	12 th of August, 2022			
HAOM BUILDERS		HSSDP		
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ISSUE 1:	For Client Approval			

CONTENTS	
1. INTRODUCTION AND PURPOSE	4
1.1 Introduction	4
1.2 Purpose	4
1.3 Structure	4
2. SCOPE OF WORKS	5
3. ENVIRONMENTAL MANAGEMENT AND MONITORING DURING CONSTRUCTION	6
4. MANAGEMENT RESPONSIBILITIES	10
4.1 Environmental/Site Manager	10
4.2 Back Up Person	11
4.3 Other Persons	11
5. LEGAL REQUIREMENTS	11
6. SPECIAL ENVIRONMENT OR CULTURAL ISSUES	11
7. MACHINERY AND SUPPORT EQUIPMENT BROUGHT TO SITE	11
8. DETAILS OF SITES USED TO SOURCE RAW MATERIALS	12
8.1 Gravel	12
8.2 Water	12
8.3 Builders Mix	12
9. CONTRACTORS FACILITIES AND WORKERS CAMPS	13
10. ENVIRONMENTAL PROTECTION WORK PROCEDURES	15
10.1 Site Preparation	15
10.2 Excavation of Construction Sites	15
10.3 Removal and Disposal of Excavated Soil	15
10.4 Erosion and Sedimentation	16
10.5 Noise and Vibration	16
10.6 Dust Generation	17
10.7 Public Access to Site	17
10.8 Mitigation of Risks to Public and Worker Health Safety and Environment (OHSE)	18

10.9	Mitigation of Spread of Covid -19 on Site	18
10.10	Usage and Storage of Hazardous Materials	18
10.11	Disposal of Waste Material (Solid and Liquid)	19
10.12	Archaeological Discoveries	19
10.13	Rehabilitation of Construction Sites and Demobilizing of Contractor Facilities	20
11.	MONITORING OF WORK	20
12.	STAFF AND WORKER TRAINING	21
13.	GRIEVANCE REDRESS MECHANISM	21
14.	SITE SET UP LAYOUT PLAN	23
15.	APPENDIX NO: 01 PHOTOS OF MACHINERY & SUPPORT EQUIPMENT THAT WILL BE BROUGHT TO SITE	24
16.	APPENDIX NO: 02 SAMPLE OF TRADE SPECIFIC METHOD STATEMENT NO: ONE (01)	26
17.	APPENDIX NO: 03 PROJECT SITE LOGBOOK	31
18.	APPENDIX NO: 04 COVID 19 MITIGATION AND MANAGEMENT POLICY	33
18.1	INTRODUCTION	34
18.2	RISK ASSESSMENT & MANAGEMENT PLAN	35
18.3	PLANNING	37
18.3.1	Recruitment & Repatriation	37
18.3.2	Site Specific Covid Response Team	37
18.3.3	Covid 19 Facilities on Site	37
18.3.4	Re-evaluate Risk Frequencies	37
18.3.5	Continuity of Work on Site	38
18.3.6	Site Operation Procedures	38

1. INTRODUCTION AND PURPOSE

1.1 Introduction

This Contractor's Environmental Management Plan (CEMP) details the implementation of measures in accordance with environmental commitments outlined in Environmental Management Plan (EMP). It will be is a contractual document which will be used by the Client as well as the Contractor (Haom Builders) to ensure the construction works complies with all the laws and regulations stated on Page 14 and 15 of the EMP. The personals involved in preparation of the CEMP are as follows;

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1.2 Purpose

The purpose of this CEMP is to outline how Haom Builders will avoid, minimize or mitigate the effects of construction work on the environment and the surrounding Agevairu community area while undertaking the project. This CEMP will

- ✚ Highlight stakeholder requirement
- ✚ Ensure that the development is compliant with the current legislations
- ✚ Outline the EMP requirements in accordance with the laws and regulations
- ✚ Detail the mitigation committed to within the EMP and how it will be implemented on site
- ✚ Ensure that any adverse effects are minimized during construction
- ✚ Describe any site-specific method statement required

1.3 Structure

This CEMP will be site specific as it will be focused on the environmental and geographical conditions of the site. This document is structured as follows

- ✚ *Scope of works* - which will define the project and provide general information in regards to the proposed development
- ✚ *Management responsibilities* - to identify the persons responsible for undertaking the work on behalf of the contractor
- ✚ *Legal Requirements* – states the laws, regulations and statutory requirements which safeguards the environment
- ✚ *Licenses and permits* – which identifies and includes consents, permission and commitments
- ✚ *Special Environmental or Cultural Issues* – identifies and locates protected areas and environment
- ✚ *Machinery and Support Equipment Brought to site* – states as declares the type of plants and equipment to use for the project which will more likely have an impact on the surrounding environment on site
- ✚ *Details of Sites Used to Source raw materials* – identifies and locates the sites which will be used for extraction of raw materials such as river gravel, water supply, timber etc
- ✚ *Contractors Facilities and Workers Camps* – details how Haom Builders will establish the site in regards to workers accommodation, amenities, storage facilities
- ✚ *Environmental Protection Work Procedures* – declares the procedures designed to minimize the impact of construction activities to the environment
- ✚ *Monitoring of Work* – provides details on how each activity will be monitored on site
- ✚ *Staff and Worker training* – states the details of training and awareness on compliance of CEMP.

2. SCOPE OF WORKS

The scope of works for the project will include the following construction activities;

- + Mobilization to Site;
- + Site Establishment;
- + Site Clearance and bulk earthworks;
- + Construction of the following facilities;
 - OPD and A&E Building
 - Outpatient Ablution Block
 - Generator Block with Solar
 - Administration Block Building
 - Guard Post
 - Inpatient Ablution Block
 - 3 Bedroom Staff House
 - Mothers Waiting House
 - Guardians Amenities
 - Kitchen & Laundry Building
 - Maintenance Workshop
 - Incinerator Block
 - TB Patient Transit House
 - Family Support Centre
 - Mortuary
 - Kuk House
 - Haus Win
 - External Works

Generally, all the above listed facilities will incorporate the following trade works;

- Ground Works – Site Works, Detail Excavation and Filling, Surface Maintenance & Preparation, Termite Treatment, Moisture Membranes
- Concrete Works – Formwork, Reinforcement, Concreting, Concrete Curing
- Stone Works – Formwork, Reinforcement, Concreting, Stone pitching
- Masonry Works – Reinforcement, Block laying, Concreting, Reinforcements
- Structural Steel Works – Posts, Beams, Painting
- Metal Works – Partition wall frames
- Carpentry & Joinery Works – Fixed Furnitures, Timber Frames and Finishes
- Hardware
- Roof and Roof Plumbing
- Ceilings
- Finishes
- Painting
- Hydraulics Services
- Drainage
- Electrical Services

3. ENVIRONMENTAL MANAGEMENT AND MONITORING DURING CONSTRUCTION

WORKING TRADES	ACTIVITIES	WORK AREA	POTENTIAL IMPACT	MITIGATION & MANAGEMENT PLAN OBJECTIVE (Design Feature/Specific Measures)	MONITORING FREQUENCY	RESPONSIBILITY
Ground Works	Site clearance	Over entire Site	Loss of existing environment (flora and fauna)	Site clearance shall be carried out within site boundary as per agreed survey plan.	During site clearance	Site Engineer/Manager
			Dry vegetation may catch fire and burn existing vegetation	Excavate pit, collect all dried vegetation and place in pit and bury them with excavated material.		
				Debranch all trees and log and pile them for later use		
				Barricade area and place fire warning signs		
	Bulk excavation and filling (cut & fill)		Affect soil stability resulting in soil erosion	Establish erosion and sedimentation control measures such as erosion or silt control/catchment sumps at low points of site.	During bulk excavation and filling	
				Batter edges of excavation and compact surface immediately after excavation to minimize loose soil		
			Affects natural drainage and waterways	Excavate diversion drains away from water ways		
			Affects water quality			
		Blockage of drainage with loose materials	Avoid blocking drainage channels with materials			
		Generation of dust during compaction	Water down site to consolidate			
		Detail excavation Detail filling, Surface maintenance, and Surface preparation	Building pads and foundations of all buildings, tank stands, septic, fence posts, solar lights, underground services, retaining wall foundations	Affect soil stability resulting in soil erosion		
	Batter edges of excavation and compact surface immediately after excavation to minimize loose soil					
	Blockage of drainage with loose materials			Avoid blocking drainage channels with materials		

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				Avoid stockpiling of materials near drains and silt containment sumps.			
			Generation of dust during compaction	Water down work area to consolidate			
	Soil Treatment	Building pads and foundations of all buildings, tank stands, septic, fence posts, solar lights, underground services	Chemical may spill or seep into drainage and water ways thus causing contamination	Avoid mixing of chemical on site Avoid application/spraying of chemicals during windy days.	During execution of Soil Treatment	Site Engineer/Manager Site Supervisor and Soil Treatment Contractor	
Concrete Works	Formwork	Foundations of all buildings, tank stands, septic, fence posts, solar lights, retaining walls, masonry walls	Saw dust may be blown into water ways	Work in an enclosed workshop of avoid spread of saw dust and noise Store all saw dust in containers and dispose as instructed by SS	During formwork construction	Site Engineer/Manager Site Supervisor, carpenters and Steel Fixers	
	Reinforcement		Rust steel can contaminate water ways	All reinforcement fixing shall be done in workshop and installed in each working area.	During steel fixing		
			Sharp objects can cause harm	Off cuts shall be stored in containers/skips for proper disposal later as instructed by SS			
	Concreting		Wet cement may seep into water ways and drains polluting existing environment from mixing and transporting of wet cement including washing and cleaning of equipment	Ensure no spills during mixing and transporting of wet concrete. Wheelbarrows shall be watertight. A containment pond shall be dug with drain connecting pond from site of mix. Transportation of wet concrete over long distances within site shall be minimized.	During concrete mixing, transporting and placing	Site Engineer/Manager Site Supervisor, carpenters, concrete finishers and casuals	
	Concrete Curing	Water from fresh concrete surface may seep into water ways and drains polluting existing environment	Minimize use of excessive water by dipping burlap mats into water skips before placing on fresh concrete rather than hosing surfaces.	During Curing			
	Stone Works	Excavation	Retaining stone walls	Refer to Ground Works		During Stone Works	Site Engineer/Manager Site Supervisor, carpenters, concrete finishers and casuals
		Backfill and Hardcore					
Formwork		Refer to Concrete Works					
Reinforcement							
Concreting							
Stone Pitching		Extraction of raw		Stones shall be collected along river			

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			materials may cause instability in river beds thus causing erosion	banks or from local quarry on dry land		
			Dirt caused by extraction may affect water quality			
Masonry Works	Reinforcements	Foundation courses, reinforced concrete block walls on specific buildings	Refer to Concrete Works		During Masonry Works	Site Engineer/Manager Site Supervisor, block layers and casuals
	Concreting					
	Block laying		Blockage of drains or water ways by buildup of broken bricks and dry mortar	All waste shall be stored in rubbish skips for proper removal and disposal as instructed by SS		
			Waste mortar or washing and cleaning of equipment may pollute water ways	Ensure no spills during mixing and transporting of wet concrete. Wheelbarrows shall be watertight. A containment pond shall be dug with drain connecting pond from site of mix. Transportation of wet concrete over long distances within site shall be minimized.		
			Dust from cutting concrete blocks may pollute surrounding environment and settle in drinking water.	All cutting of concrete blocks shall be done in an enclosed area. Dust shall be minimized by apply water during cutting.		
Structural Steel Works	Fabrication	Posts, King Posts, floor beams and other structural framing for all buildings	Fabrication will be done off site so no potential harm to the environment will be present.		During Structural Steel Works	Site Engineer/Manager Site Supervisor,Welders/Riggers
	Installation		Installation will be done on site; however, this activity will not pose any threat to the environment.			
Metal Works	Fabrication	Wall frames, ceiling frames, roof frames, security screens, veranda rails and uprights on specific buildings	Fabrication will be done off site so no potential harm to the environment will be present.		During Metal Works	Site Engineer/Manager Site Supervisor, Carpenters
	Installation		Rust steel can contaminate water ways	All off cuts shall be stored in rubbish skips and buried at a specific location		
			Sharp objects may injure people			
Carpentry & Joinery Works	Fabrication	Wardrobes, shelves, benches, cabinets, timber finishes other	Fabrication will be done off site so no potential harm to the environment will be present.		During Carpentry & Joinery Works	Site Engineer/Manager Site Supervisor, Joiners
	Installation		Installation will be done on site; however, this activity will not pose any threat to the environment.			

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		timber works on specific buildings				
Hardware	Installation	Door hardware, window hardware, sanitary hardware and general hardware on all buildings	Installation will be done on site; however, this activity will not pose any threat to the environment.		During Hardware installations	
Roof and Roof Plumbing	Installation	On all buildings	Rust steel can contaminate water ways	All roofing materials shall be supplied to measurement details. All off cuts shall be stored in rubbish skips and buried at a specific location	During Roof and Roof Plumbing Installation	Site Engineer/Manager Site Supervisor, Plumbers
			Sharp objects may injure people			
Ceilings	Framing installation	On specific buildings	Sharp objects may injure people	All off cuts shall be stored in rubbish skips and buried at a specific location	During Ceiling Installation	Site Engineer/Manager Site Supervisor, Carpenters
	Lining installation		Dust and off cuts can cause harm or contaminate environment	Cutting shall be done in enclosed area		
Finishes	Plastering		Chemicals and residues may seep into water ways	Avoid spills during mixing and application.	During Finishes	Site Engineer/Manager Site Supervisor, Carpenters. Tilers and Plasterers
	Vinyl Tiles and Ceramic Tiles		Saw dust can affect air quality	Work in an enclosed area or workshop and install.		
	Timber Finishes		Chemicals and residues may seep into water ways	Avoid spills during mixing and application.		
	Wall rendering					
Painting	1 st Coats	On all buildings	Chemicals and residues may seep into water ways	Avoid spills during mixing and application.	During Painting	Site Engineer/ Manager Site Supervisor, Painters
	2 nd Coats					
Hydraulic s Services	Rough-ins	On specific buildings	Sharp objects may injure people	All off cuts shall be stored in rubbish skips and buried at a specific location	During Hydraulics Services	Site Engineer/ Manager Site Supervisor, Plumbers
	Fit outs		All fit outs will be done within each of the buildings			
Drainage	Excavate Trench	On specific buildings	Refer t Ground Works - Excavation		During Drainage	
	Install pipes		Offcuts may pollute environment	All off cuts shall be stored in rubbish skips and buried at a specific location		
Electrical Services	Rough-ins	On all buildings	Offcuts may pollute environment			During Electrical Services
	Fit outs					

4. MANAGEMENT RESPONSIBILITIES

The persons identified in the following tables will be assigned and attached to form the site management team for the Construction of Agevairu Health Centre Stage I & II Project. They will be directly responsible for the CEMP activities. The Site Manager is the most superior person on site followed by the Site Supervisor. He will ensure that all the works and activities on site conform to the CEMP. The organizational chart on the following page illustrates the management structure, the proposed channel of communication and authority on management of the CEMP on site.

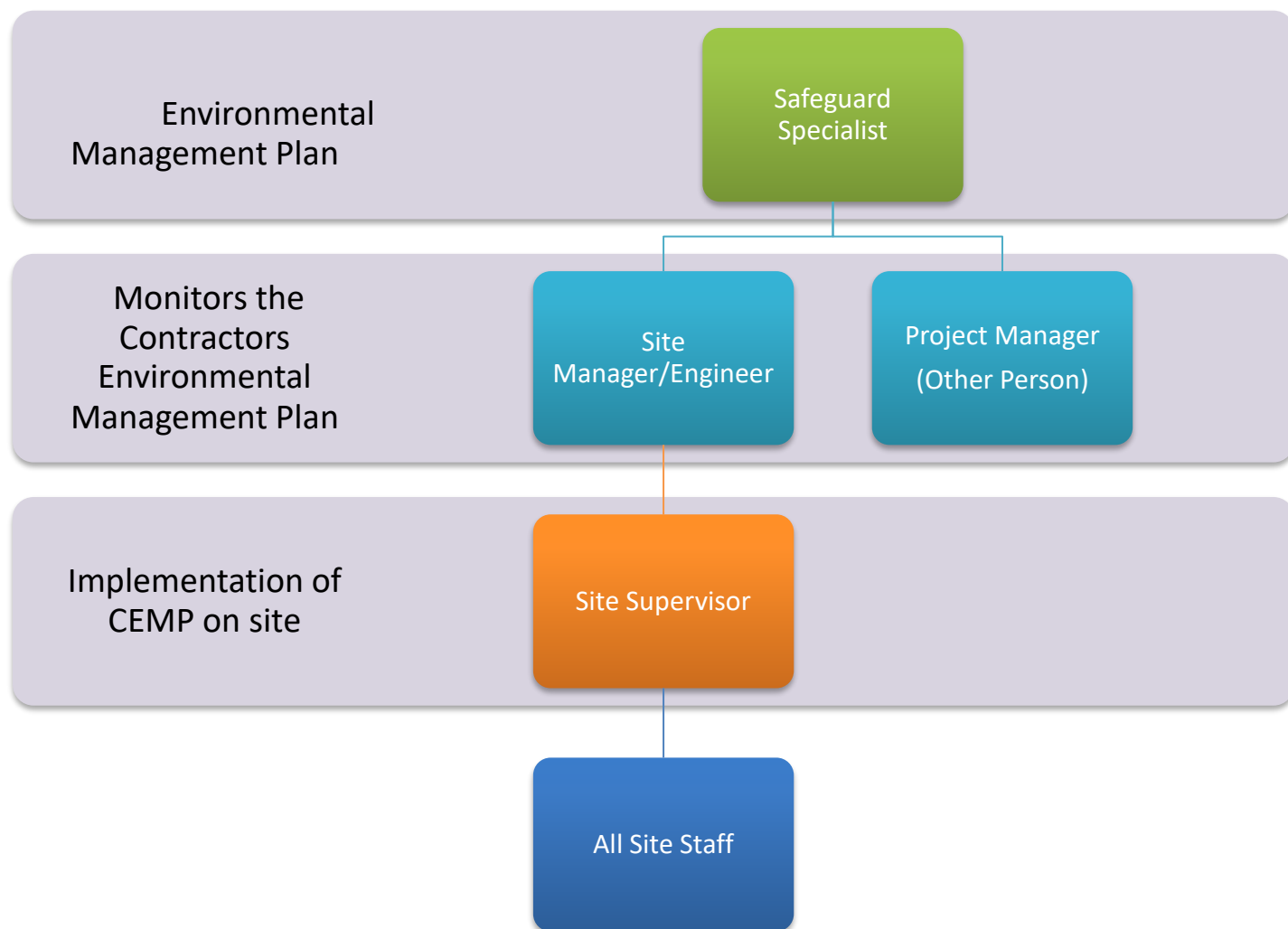


Figure 1. Organizational Structure for the onsite management of CEMP

These persons may be replaced by other persons after submission of this document. The officer in charge will be made aware of such changes at the earliest juncture.

4.1 Environmental/Site Manager

Name	Position	Availability on site	Office	Mobile	Email
Mr. Jeremiah Rocky	Environmental / Site Manager	Full Time	472 0601	71742309	haombuilders@gmail.com

4.2 Back Up Person

Name	Position	Availability on site	Office	Mobile	Email
Mr. Francis Ero	Site Supervisor	Full Time	472 0601	72322660	haombuilders@gmail.com

4.3 Other Persons

Name	Position	Availability on site	Office	Mobile	Email
Mr. Hamo Anep	Projects Manager	Part Time	472 0601	72322660	haombuilders@gmail.com

5. LEGAL REQUIREMENTS

We, Haom Builders, understand that it is our responsibility to ensure that all the works carried out on site is within the jurisdiction of various environmental laws, regulations and standards mentioned in this CEMP. We understand that we will always seek clarification from these documents should any conflict arise during the course of work. We acknowledge the following legal documents to form part of the legal requirements along with this CEMP;

- ✚ ADB Safeguard Policy Statement
- ✚ Environmental Act 2000
- ✚ Environmental Prescribed Activities Regulations
- ✚ Land Group Incorporation Act 1974
- ✚ Land Registration Act 1981

6. SPECIAL ENVIRONMENT OR CULTURAL ISSUES

Local knowledge will be sought if we come across anything of cultural significance is located on site during the course of work. As per the EMP, the river is 400 east of the site. Streams and water wells are some distance away from the site. However, measures will be taken to prevent spills of pollutants into the lake or any stream that runs into the streams. Plants and vegetation immediately outside the perimeter of the site will be protected as much as possible. All surface run offs will be diverted into soak up pits on site. Contaminated spills and liquid wastes will also be disposed in separate containment/soak up pits.

River gravel for filling and surface improvement on site will be sourced from the quarry opposite the site after consultation with local land owners. If local suppliers will supply, then they will be briefed on Environmental Safeguards requirements under this contract so that they minimized environmental damage while extracting materials.

Existing buildings will be protected from being damaged. Barricades will be put in place to identify the boundary of the building. Compaction of the main site areas will be addressed thoroughly to prevent disruption to existing buildings foundation.

7. MACHINERY AND SUPPORT EQUIPMENT BROUGHT TO SITE

The following table shows the details of machineries and equipment that we intend to commit to the project. Please refer to Appendix No: 02 for some of the photos of these equipment.

PLANT NAME	MAKE & TYPE	REGO	AGE	CONDITION	CURRENT LOCATION
1x 3m ³ Tonne Dump Truck	Isuzu	LBF 211	4 yrs old	Recently serviced and running – Very Good Condition	Lae, Morobe Province
1x Land Cruiser (Utility)	Toyota	LBF 590	4 yrs old	Recently serviced and running – Very Good	Lae, Morobe Province

			Condition	
2x 5KVA Generator	Powertrade	Brand New	Very Good Condition	Lae, Morobe Province
2x 300L concrete mixers	Truemax	Brand New	Very Good Condition	Lae, Morobe Province
1xConcrete vibrator	Truemax	Brand New	Very Good Condition	Lae, Morobe Province
30x Scaffold set		Brand New	Very Good Condition	Wabag, Enga Province
2x Portable welding machine – 175 inverter	Weldmaster	1 year old	Very Good Condition	Wabag, Enga Province
1xAutomatic Dumpy Level	Sokkia	1 year old	Very Good Condition	Wabag, Enga Province
1xTamping Rammer	Truemax	6 months old	Very Good Condition	Lae, Morobe Province
1xVibratory Plate Compactor	Truemax	6 months old	Very Good Condition	Lae, Morobe Province

8. DETAILS OF SITES USED TO SOURCE RAW MATERIALS

8.1 Gravel

As stated, earlier gravel will be sourced from local quarry. This gravel will be used for improving surface condition of soil. Access to the is in good condition and is the only gravel extraction point along that portion of the road. Civil contractors normally get gravel for their road projects in the area.

8.2 Water

Water for construction work will be piped separately from the existing well. Haom Builders intends to fix the well and get water and store in tanks and supplied by gravity feed. Existing water supply to the current health center will not be disturbed unless permission is granted. Drinking water will be stored in tanks and rainwater also collected into tanks for various usages. Additional water for construction works can be pumped from the nearby creeks if necessary or during dry season.

8.3 Builders Mix

Builder's mix will be sourced from suppliers within Port Moresby or suppliers along Hiritano Highway and transported to site. We will utilize our dump truck to transport these builders mix to the site.

9. CONTRACTORS FACILITIES AND WORKERS CAMPS

The Table below contains the general scope of site facilities. Please note that the floor areas of the facilities may decrease due to space limitations or increase due to functional requirement on site. Appendix No: 01 shows the location of these facilities on site.

FACILITY	AREA (m²)	CONSTRUCTION			STORM WATER DRAINS TO:	EFFLUENT DRAINS TO:	ELECTRICITY SUPPLY SOURCE	COLD WATER SOURCE
		FLOOR	WALLS	ROOF				
Site Office - Project manager's office - Site Engineer's Office - Meeting Room - Toilet - Tea room	24m² (6m long x 4m wide)	Timber framed on low post, 18mm ply boards	Timber framed with V-crimp to external walls & 4mm plywood to internal walls	Timber framed with corrugated roofing sheets	Sediment basin (natural drainage system)	5000L Tuffa Plastic Septic Tank	5kva Genset	9000L TUFFA Tank & existing supply
Site Manager's Accommodation & Site Engineers Accommodation	16m² (4m long x 4m wide)	Timber framed on low post, 18mm ply boards	Timber framed on low post, 18mm ply boards	Timber framed with corrugated roofing sheets	Timber framed with corrugated roofing sheets	5000L Tuffa Plastic Septic Tank		9000L TUFFA Tank & existing supply
Workers Accommodation	90m² (15m long x 6m wide)	Compacted river gravel base/floor	Timber framed with V-crimp to external walls & 4mm plywood to internal walls	Timber framed roof with corrugated roofing sheets	Sediment basin (natural drainage system)			
Workers Kitchen & messing facilities	36m² (9m long x 4m wide)	Compacted river gravel base/floor	Timber framed with V-crimp to external walls & 4mm plywood to internal walls	Timber framed roof with corrugated roofing sheets	Sediment basin (natural drainage system)			
Shower rooms and laundry area	18m² (6m long x 6m wide)	Reinforced Concrete floor	Timber framed with V-crimp to external walls & 4mm plywood to internal walls	Timber framed roof with corrugated roofing sheets	Sediment basin (natural drainage system)			
Material & Small plants &	100m² (10m square)	Compacted river gravel	Timber framed with V-crimp to	Timber framed roof with	Sediment basin (natural drainage	NA		NA

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Equipment Storage Shed		base/floor	external walls & 4mm plywood to internal walls	corrugated roofing sheets	system)			
Visitors Toilet - Male - Female	8m2 (4m x 2m)	Timber framed on low post, 18mm ply boards	Timber framed on low post, 18mm ply boards	Timber framed with corrugated roofing sheets	Timber framed with corrugated roofing sheets	5000L Tuffa Plastic Septic Tank		9000L TUFFA Tank & existing supply
Vehicle and plants lay down area	100m ²	Compacted river gravel base/floor	NA	NA	Sediment basin (natural drainage system)	NA	NA	NA

10. ENVIRONMENTAL PROTECTION WORK PROCEDURES**10.1 Site Preparation**

- ✚ Top soil will be altered by means of mechanical excavation
- ✚ Exposed surface of soil may be subjected to erosion and sedimentation at lowest points.

Objective	To appropriately plan and prepare site so that damage to environment on site is minimum.		
Actions	Requirements	Responsibility	Timing
	No marine or terrestrial vegetation will be disturbed without prior approval		Throughout construction works
Performance Indicators	No injury / harm to local flora and fauna	Site Manager, Site Supervisor	Throughout construction works
Monitoring	Visual monitoring		Throughout construction works
Reporting	Reporting to Site Manager and SS		Throughout construction works
	Any reporting to regulators		
Corrective Actions	Corrective actions and implementation time frame		Throughout construction works

10.2 Excavation of Construction Sites

- ✚ Potential sources of sediments
- ✚ Risks of erosion
- ✚ Stockpiles
- ✚ Stormwater discharge points
- ✚ Existing field and kerb inlets and other stormwater infrastructure
- ✚ Site access and egress locations
- ✚ Nearby sensitive areas

Objective	To manage construction activities which exhibit the potential to cause erosion and release of sediment		
Actions	Requirements	Responsibility	Timing
	Minimize erosion and water ponding Minimize injuries to workers		Prior to construction commencing
Performance Indicators	No mud/dirt visible on construction site		Throughout construction works
Monitoring	Daily visual inspections of works site and all erosion and sediment controls	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Central Provincial Health Authority	SS	
Corrective Actions	Corrective actions and implementation time frame	Site Manager, Site Supervisor	Throughout construction works

10.3 Removal and Disposal of Excavated Soil

Include:

- ✚ All excavated materials will be reused to fill up lower areas and retaining walls
- ✚ Top soil will be stored and used for landscaping works

Objective	To appropriately manage excavated materials over site and minimize disposal off site.		
Actions	Requirements	Responsibility	Timing
	Suitable location will be identified for storage of excavated materials.	Site Manager	Prior to works commencing

CONTRACTORS ENVIRONMENTAL MANAGEMENT PLAN (CEMP)

	Materials will be separated and reused appropriately		
Performance Indicators	Minimize unnecessary stockpile of materials and erosion of materials	Site Manager	Throughout remediation works
Monitoring	Daily inspection of storage area	Site Manager	Throughout remediation works
Reporting	Reporting to Site Manager and SS Central Provincial Health Authority	All staff SS	Throughout construction works
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

10.4 Erosion and Sedimentation

Include:

- ✚ Potential sources of sediment/risks of erosion
- ✚ Stockpiles
- ✚ Stormwater discharge points in open drains and soak up pits
- ✚ Existing field and kerb inlets and other stormwater infrastructure
- ✚ Site access and egress locations
- ✚ Nearby sensitive areas
- ✚ Diversions for surface run offs

Objective	To manage construction activities with the potential to cause erosion and the release of sediment		
Actions	Requirements	Responsibility	Timing
	Construct sumps or absorption pits along certain points of drains to keep eroded materials	Site Manager	Prior to construction commencing
Performance Indicators	Minimize erosion over site	Site Manager	Throughout construction works
Monitoring	Daily visual inspections of works site and all erosion and sediment controls	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS Central Provincial Health Authority	All staff SS	Throughout construction works
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

10.5 Noise and Vibration

Include:

- ✚ Noise generating activities such as steel fabrication will be done off site.
- ✚ Maintenance and operation of machinery / equipment / vehicles
- ✚ Work hours
- ✚ Specify any activities to be undertaken at night and why (if any)

Objective	To appropriately manage noise during construction activities to minimize impact to workers, neighbors and community members.		
Actions	Requirements	Responsibility	Timing
	all significant noise generating activities will be undertaken between Mon-Sat 6.30am to 6.30pm	Site Manager	Throughout construction works
Performance Indicators	no complaints regarding noise from workers / neighbors / community members	Site Manager	Throughout construction works
Monitoring	noise monitoring program or complaint based monitoring	Site Manager	Throughout construction works

CONTRACTORS ENVIRONMENTAL MANAGEMENT PLAN (CEMP)

Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Central Provincial Health Authority	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

10.6 Dust Generation

- ✚ Include the following:
- ✚ Dust suppression
- ✚ Stockpiles
- ✚ Truck loads
- ✚ Maintenance and operation of machinery / equipment / vehicles
- ✚ Other methods to prevent / minimise dust

Objective	To manage construction activities with the potential to impact on air quality		
Actions	Requirements	Responsibility	Timing
	water carts will be used for dust suppression	Site Manager	Throughout construction works
Performance Indicators	no complaints regarding dust from workers / neighbors / community members	Site Manager	Throughout construction works
Monitoring	air quality monitoring program or complaint-based monitoring	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Central Provincial Health Authority	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

10.7 Public Access to Site

Include the following:

- ✚ Installation of proper signage barricades
- ✚ Logbooks and entry passes
- ✚ Security at all entry and exit point

Objective	To manage construction activities with the potential to prevent theft and vandalism, and accidents or injuries to all workers, contractors and visitors		
Actions	Requirements	Responsibility	Timing
	Site will be accessed via guard house	Site Manager	Throughout construction works
Performance Indicators	No trespasser on site.	Site Manager	Throughout construction works
Monitoring	Security checks at entry points	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Central Provincial Health Authority	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

10.8 Mitigation of Risks to Public and Worker Health Safety and Environment (OHSE)

Include the following:

- ✚ Provide PPEs;
- ✚ PPE checks;
- ✚ Truck loading checks and requirements;
- ✚ Safety warning signs;
- ✚ Barricades and danger tapes;
- ✚ Maintenance and operation of machinery / equipment / vehicles;
- ✚ Other methods to prevent / minimise dust;
- ✚ Chemical spill containments;
- ✚ Traffic warning signs on Kerema Highway.

Objective	To manage construction activities with the potential to impact Occupational Health Safety and Environmental Damage		
Actions	Requirements	Responsibility	Timing
	All workers will be to have all necessary PPEs	Site Manager	Throughout construction works
Performance Indicators	No injuries or accidents on site	Site Manager	Throughout construction works
Monitoring	PPE check every morning during toolbox meeting	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Central Provincial Health Authority	SS	
Corrective Actions	Corrective actions and implementation time frame	Site Manager	Throughout construction works

10.9 Mitigation of Spread of Covid -19 on Site

Include the following:

- ✚ Provide hand washing stations on all entry points;
- ✚ Provide hand sanitizers in site offices;
- ✚ Practice social distancing during all meetings on site;
- ✚ Provide Covid – 19 mitigation signs;
- ✚ Implement Covid – 19 Mitigation Policy;
- ✚ Carry out awareness on site and surrounding community.

Objective	To manage and mitigate spread of Covid – 19 within construction site which may affect construction works		
Actions	Requirements	Responsibility	Timing
	All workers will be to have all follow all mandatory protocol or new normal anywhere on site.	Site Manager	Throughout construction works
Performance Indicators	No spread of Covid – 19 on site	Site Manager	Throughout construction works
Monitoring	Temperature checks and regular hand washing and sanitation	Site Manager	Throughout construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Central Provincial Health Authority	SS	
Corrective Actions	Corrective actions and implementation time frame	Site Manager	Throughout construction works

10.10 Usage and Storage of Hazardous Materials

- ✚ Storage

CONTRACTORS ENVIRONMENTAL MANAGEMENT PLAN (CEMP)

- ✚ MSDS's
- ✚ Refuelling
- ✚ Spill kits and emergency spill response
- ✚ Spill kit training
- ✚ Wrapping or covering of powdered materials (cement, grouts etc)

Objective	To manage construction activities with the potential to cause contaminated soil and harm workers		
Actions	Requirements	Responsibility	Timing
	All hazardous will be stored in secured, roofed areas. Use of canvas or Visqueen, sawdust or gravel bedding for fuel storage/refueling areas to be replaced if spillage is encountered.	Site Manager	Throughout construction works
Performance Indicators	No spills or injuries to workers	Site Manager	Throughout construction works
Monitoring	chemical storage areas and spill kits will be inspected on a routine basis	Site Manager	Throughout construction works
Reporting	Central Provincial Health Authority	All staff	Throughout construction works
	Central Provincial Health Authority	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

10.11 Disposal of Waste Material (Solid and Liquid)

Include:

- ✚ Waste burial pits organic materials
- ✚ Disposal bins for inorganic materials

Objective	To appropriately manage waste during construction activities.		
Actions	Requirements	Responsibility	Timing
	Appropriate waste receptacles will be provided on site	Site Manager	Prior to works commencing
Performance Indicators	No waste deliberately or unintentionally released	Site Manager	Throughout remediation works
Monitoring	Reporting to Site Manager and SS	All staff	Throughout remediation works
Reporting	Central Provincial Health Authority	SS	Throughout construction works
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

10.12 Archaeological Discoveries

Include:

- ✚ Any known areas of Cultural Heritage Significance
- ✚ Steps to be undertaken if cultural heritage artefact is uncovered ("cultural heritage duty of care")

Objective	To appropriately manage construction activities with the potential to disturb / uncover cultural heritage sites / Artifacts.		
Actions	Requirements	Responsibility	Timing
	All workers will be informed of anything of cultural significance on site	Site Manager	Throughout construction works
Performance Indicators	No cultural heritage artifacts uncovered or damaged	Site Manager	Throughout construction works
Monitoring	Visual monitoring	Site Manager	Throughout

			construction works
Reporting	Reporting to Site Manager and SS	All staff	Throughout construction works
	Any reporting to regulators	SS	
Corrective Actions	corrective actions and implementation time frame	Site Manager	Throughout construction works

10.13 Rehabilitation of Construction Sites and Demobilizing of Contractor Facilities

Include:

- ✚ Schedule maintenance of site
- ✚ Housekeeping over site
- ✚ Removal of contaminated soil
- ✚ Removal of solid wastes

Objective	To appropriately manage construction site so that minimum damage is done to environment close to the proximity of site.		
Actions	Requirements	Responsibility	Timing
	Maintenance of site facilities and housekeeping during construction period.	Site Manager	Throughout construction works
Performance Indicators	Well organized and kept site with no rubbish or contaminants on soil	Site Manager	Throughout construction works
Monitoring	Visual monitoring	Site Manager	Throughout construction works
Reporting	Reporting to site manager and SS	All staff	Throughout construction works
	Any reporting to regulators	SS	
Corrective Actions	Corrective actions and implementation time frame	Site Manager	Throughout construction works

11. MONITORING OF WORK

Monitoring of CEMP will be done progressively on a daily basis. Two (02) documents will be used to with the CEMP to monitor environmental issues on site. The CEMP will be used together with our Trade Specific Method Statement and Project Site Diary. Samples of each of these documents are attached in Appendix No. 03 and No: 04. The method statement will outline how each activity in each of the working trades will be carried out while considering potential risks to both the environment and the progress of work itself. The method statement will be issued to the Safeguard Specialist (SS) before commencement of work.

For example, if we are to commence concrete works after the completion of groundwork's, a fully furnished method statement will be issued a week prior to the commencement of work on a particular trade. The method statement will state how all the activities in this trade will be carried out in sequence. Considerations will be given to the nature of impact each of the activities will have on the environment. This potential impact will be identified as a risk under Part 6 of the Trade Specific Method Statement.

The site engineer and the supervisor will ensure that the risk mitigating factors on the method statement are executed on site. The Project Manager and the Site Engineer will ensure that the SS is well informed on any changes in the method statement.

Site Log Book will also be used to record all activities on site each day. It will also have provisions where all activities relating to environmental protection impact mitigation will be recorded. The site engineer will be responsible for keeping record in the Project Site Logbook.

Monthly reports on CEMP activities will summarise all the records on the daily site dairy. It will also status of all activities in the method statement and CEMP. The Project Manager and the Site Engineer will be responsible for compilation of the report.

12. STAFF AND WORKER TRAINING

All the site personnel recruited for the Project will go through induction. They will be made aware of the CEMP requirements prior to mobilisation to site. Our standard procedure for every project site is organising a Tool Box Meeting every morning prior to work. From this tool box meeting, awareness will be made concerning various areas of concern in the CEMP.

When commencing work on new trade area, all site personnel will congregate to an assembly point to be informed about the risks and the impacts that particular trade area will have on the environment. They will be informed on waste management and rubbish disposal strategies that are in place. The project manager and the site engineer will work in collaboration with the Safeguard Specialist. Awareness may be in form of verbal explanation, pictorial illustrations, and video and power point presentations.

The CEMP reporting structure will also be made clear so that workers will know whom they will report to in terms of CEMP requirement. Monthly or weekly reports as agreed will be submitted to the SS and SO for review of the CEMP if necessary.

13. GRIEVANCE REDRESS MECHANISM

All grievances concerning the environment will be addressed by the Site Manager. However, we will also involve our managing director who is also responsible for public relations of the company in Enga Province. We will utilize the Grievance Intake Form attached in Appendix 1 in the EMP. Permission will be sought from safeguard specialist or concerned authorities.

Other complaints regarding the project itself will be handled by the Site Manager and the Public Relations Officer (PRO). Our PRO in Agevairu will be a Local Community Leader or a Church Pastor. He has been in charge of the addressing grievances in some of our past projects in Enga Province.

Complaint forms will also be provided at the site office. The project complaint form will be a different form the Grievance Intake Form. Complainants will write down their complaints on the forms. All complaints will be referred to the Site Manager and the PRO who will then assess and give feedback the way they see appropriate.

The proposed procedure in addressing complaints/grievances concerning the environment is illustrated by the process chart on the following page.

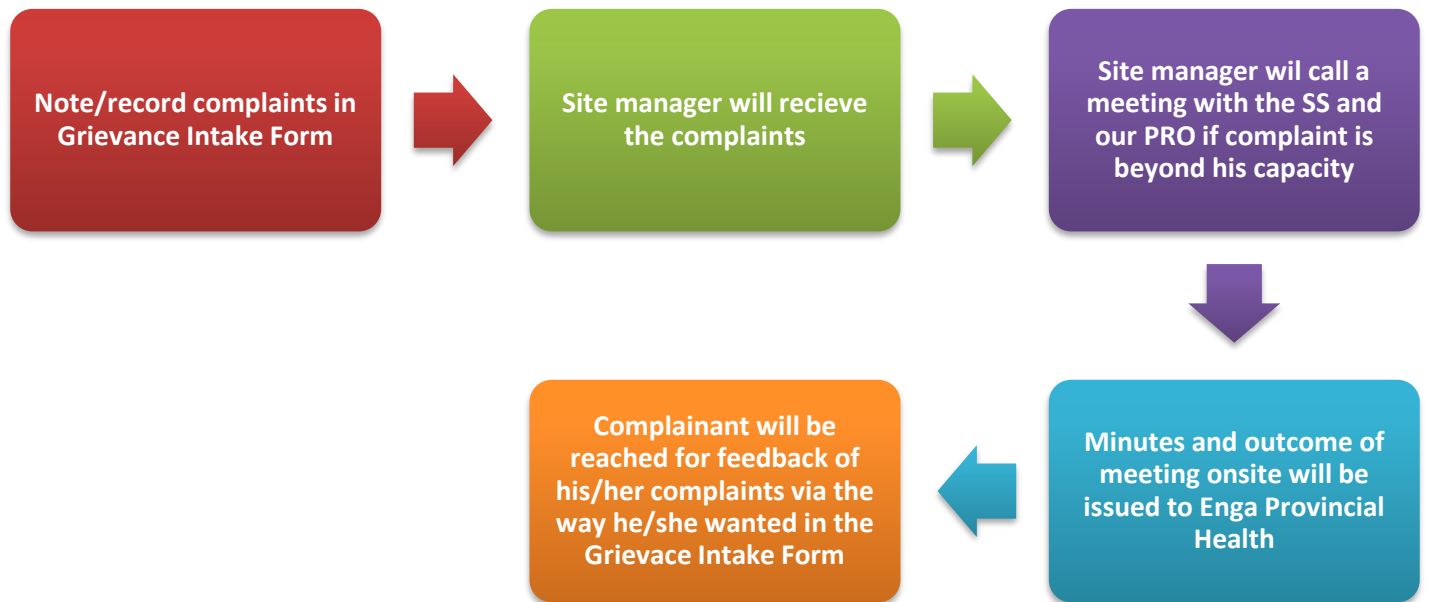


Figure 2. Steps to be taken to address grievances on site

14. SITE SET UP LAYOUT PLAN

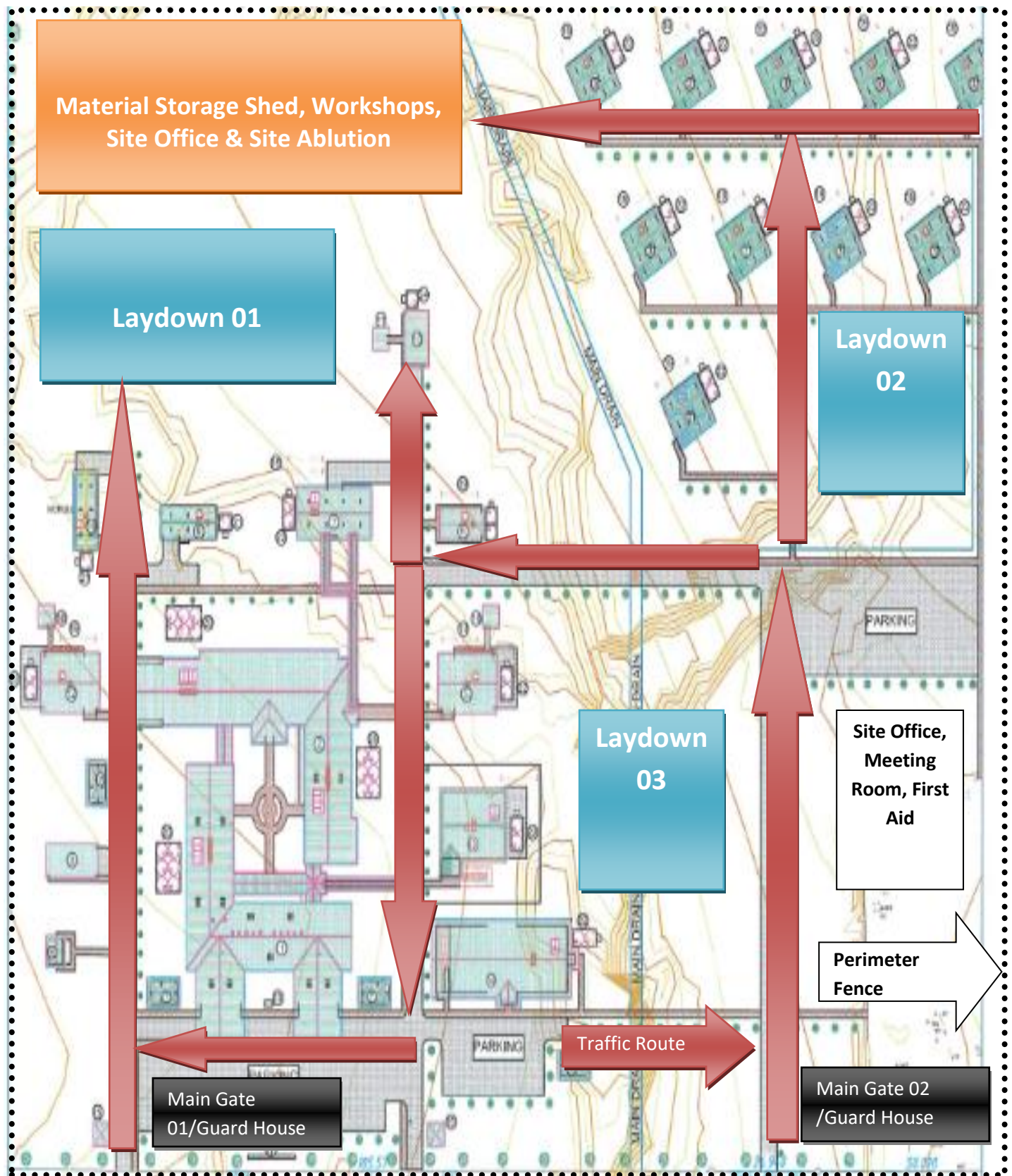


Figure 3. Proposed Site Layout

15. APPENDIX NO: 01 PHOTOS OF MACHINERY & SUPPORT EQUIPMENT THAT WILL BE BROUGHT TO SITE



3m³ Dump truck to be used for logistics of materials from lay-down yard at Wabag to sites



Toyota Land Cruiser which will be used at Porea CHP



Isuzu NPS 17 Tonne Flat Tray Truck to be used for Logistics of Materials from Lae to Wabag



One of the 5KVA generators to be used for the Porea CHP



300L cement mixers bought and ready to be brought to site.



Newly bought tamping rammer & vibratory plate compactor



Mitre radial cross cut saw for cutting timber

16. APPENDIX NO: 02 SAMPLE OF TRADE SPECIFIC METHOD STATEMENT NO: ONE (01)

PART 1: PROJECT DETAILS

TRADE SPECIFIC METHOD STATEMENT NO:		HB001	DATE	
SITE ADDRESS	Agevairu Health Centre, Kairuku-Hiri District, Central Province			
PROJECT SITE	Agevairu	PRINCIPLE	National Department of Health	
PROJECT NAME	Health Services Sector Development Project (HSSDP)			
CONTRACT NO:	PN#CW-0401-HSSDP/NCB/61/0322	CLIENT	Central Provincial Health Authority	
PROJECT MANAGERS	HSSDP			

PART 2: CONTRACTOR DETAILS

CONTRACTOR	Haom Builders	ADDRESS	P.O. Box 797, LAE Morobe Province	
PROJECT MANAGER	Mr. HAMO ANEP		QUALIFICATION	BBLD.UoT
SITE MANAGER	Mr. Jeremiah Rockey/Mr. Benjamin Igare		QUALIFICATION	BBLD.UoT
SITE SUPERVISOR	Mr. Amba Giactulu/Mr. Babayu Kerry		QUALIFICATION	Tradesman Certificate

PART 3: SCOPE OF WORKS & WORK BREAK DOWN

BUILDING ELEMENT		Substructure		BUILDING LEVEL		Ground Floor Level	
WORKS INVOLVED		Groundwork's		DURATION			
ACTIVITIES INVOLVED		MAIN MATERIAL(S)		PLANTS/EQUIPMENTS		SKILLS REQUIRED	
		DESCRIPTION	QTY	DESCRIPTION	QTY	DESCRIPTION	QTY
1	Detailed Excavation Below Ground Floor Level for: Stripped Footings Pad Footings	Excavated material	98.4 m ³	Automatic Dumpy Level & staff	2	Driver	1
				Building tape	5	Carpenters	24
				String lines 100mm	10	TA Carpenters	12
				Chalk line	10		
				Spades	20		
				Crew bar	5		
				Wheel barrows	10		
				Hydraulic rammer	2		
				Cutting blades	10		
2	Maintaining Faces		109 m ²				
3	Surface Preparation Trimming of sides, leveling of surfaces	Battered, trimmed, graded and compacted surfaces	212 m ²				
4	Hardcore filling & Compaction Placing of approved hardcore fill material & compact to required dry density.	Diorite, Road base or equally approved backfill material	188.6 m ³				
5	Soil Treatment	Subcontractor to provide materials	574 m ²				
6	Moisture Membrane	Polythene membrane	459 m ²				

SEQUENCE OF WORK

CONTRACTORS ENVIRONMENTAL MANAGEMENT PLAN (CEMP)

Excavation of foundations below ground floor level will commence once profiles area confirmed by the engineer and the Superintendent.

Profiling and setting out for excavation will be done once all site establishment works are completed. Surface to be excavated should be marked with chalk line.

All excavation will be done manually as the sizes of footings do not allow mechanical means of excavation.

Maintenance of faces after mechanical means will be additional work.

All excavated material will be stock piled 1m or 2m away from the trenches and removed by the backhoe.

Sides and faces of excavation will be trimmed and leveled. All loose soil will be removed from the surface.

Back fill materials will be transported to specific work areas using wheel barrows and placed.

Compaction will be done by hydraulic rammer to specified dry density. Compaction will be done after every 150mm.

Heights and levels will be maintained by taking levels using dumpy level and height pegs.

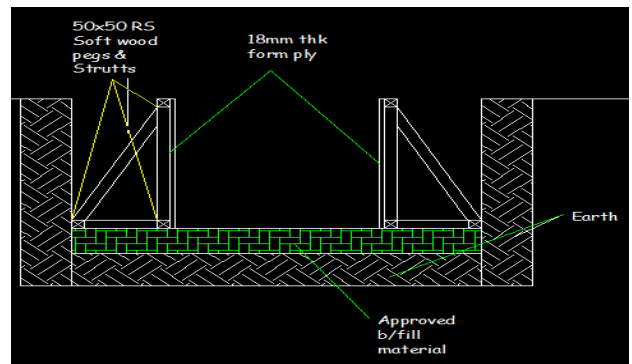
Soil treatment will be carried out by a Specialist Subcontractor.

Moisture membranes will be laid and lapped as specified. All ends will be securely fastened to the soil or other supports.

All loose soil or rubbish on the surface of moisture membranes will be removed before insertion of reinforcement.

PART 4: ADDITIONAL SUPPORT/STRUTTING

Temporary supports and propping/strutting will be constructed to prevent erosion of sides of excavation. Barricades will be erected on weak soil around the perimeter of excavation. Opposite is an example of how we intent to construct our support if we see necessary.



PART 5: METHOD OF ACCESS AND EGRESS

ACCESS TO GROUND FLOOR LEVEL

Barricades will be erected around excavated area. Entry points of small plants and employees will be established.

Traffic routes will be established away from profiles and excavation.

No employee will enter the excavated area without authorization from Engineer or Supervisor.

Heavy machines such as trucks will not be allowed close to edge footing excavation.

All materials into the foundations will be transported via wheel barrows to prevent loose edges of excavation from being damaged.

WORKING SPACES

Working spaces will not be allowed in trench excavation for footings unless there is weak soil to support. 400mm will be the minimum allowable working space.

PART 6: RISK IDENTIFICATION

POTENTIAL RISKS	POTENTIAL IMPACT ON WORK	COUNTER MEASURES
BAD WEATHER The site experiences unpredictable weather patterns with prolonged rainy season all throughout the year.	Delay in delivery of materials on site	Seek information on daily weather forecasts from Provincial Weather Office or reliable sources so that it should be considered in planning and execution of work on site.
	Water logging over site and in bottom of excavated surface.	All excavated surface of the trenches will be covered with canvas. If waterlogged, water will be pumped out using buckets or fuel generated pumps.
	Surfaces of excavation may erode & sediments may build up at the lowest point of site in front of the staff residences	An absorption pit will be excavated at the lowest point of the site to receive water and sediments from all drains.
	Work will be stopped on rainy days which may affect work progress and schedule	No electrical tools will be needed during this work. Canvas canopy will be constructed above working area so work continues. Proper drainage will be done to remove excessive water.

CONTRACTORS ENVIRONMENTAL MANAGEMENT PLAN (CEMP)

ACCIDENTS & INJURIES ON SITE An employee or a member of the public would be injured.	Work would be stopped or delayed.	Implement Haom Builders Safety Policy on site and monitor the policy on site.
		Tool box meetings will be held every morning for safety checks and briefing
		Temporary fence and hording will be erected. Entrance and exit gates will be guarded.
		Establish traffic management on site and designate routes to work areas of employees, excavation plants and delivery vehicles.
		Erect barricades and signs indicating potential danger on site.
		No vehicle or other plants routes will be located close to the edge of excavation.
		Temporary supports and strutting will be erected on weaker soil on the surface around the perimeter of excavation.
		Only qualified operators will be allowed to operate hydraulic rammers. No worker will be allowed to stand close to the machine.

PART 7: HARZARD IDENTIFICATION

HEALTH HARZARDS DURING EXCAVATION OF FOUNDATIONS ON GROUND FLOOR LEVEL		MITIGATION OF HARZARDS
1	Noise pollution from excavators and vehicles when in use.	Hoardings will be erected as a noise barrier.
		Engines of all plants should be switched off when at rest.
		Ear muffs will be issued to every employee on site
2	Risk of falling into excavation area	Hoearding will be erected around the perimeter of excavation. Temporary supports will be constructed to hold loose soil on excavated sides in place.
		No employee or unauthorized person will be allowed close to excavation or into where excavation is done.
3	Dust pollution due to moving machines and work	Dust masks, safety glasses and other PPEs will be issued to each employee on site
		Hoardings will be erected as a dust barrier.
4	Traffic on site	Implement onsite traffic management plan

PART 8: SAFETY MANAGEMENT ON SITE

SAFETY POLICY	EMERGENCY PROCEDURES	FIRST AID FACILITIES	
Haom Builders Safety policies include an Onsite Safety Management Plan which sets out how safety measure will be implemented, monitored and enforced on site. Please refer the Safety Policies and Management Plan	Please refer to Haom Builders Safety Policies and Management Plan	Safety Officer	Peter Anep
		First Aid Office Location Indication	
		Nearest Health Center with	Agevairu Health Centre
		Ambulance Services	St Johns Ambulance and Agevairu Health Centre Ambulance
		Nearest Hospital	Pom General Hospital

PART 9: PERSONNEL PROTECTION EQUIPMENT ON SITE

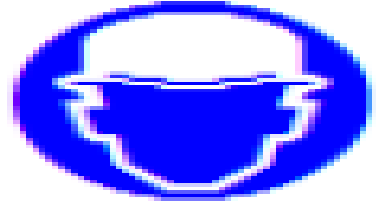
MANDATORY PERSONNEL PROTECTION EQUIPMENT TO BE USED ON SITE



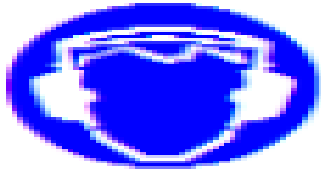
**Safety
Boots**



**Safety
Gloves**



Hard Hats



**Hearing
Protection**



**Eye
Protection**



**Respiratory
Protection**

MANDATORY COVID – 19 MITIGATION SIGNS

SAFETY FIRST



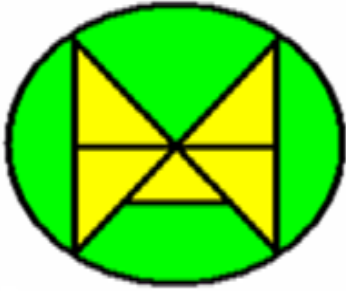
**PLEASE WEAR
FACE COVERINGS
THANK YOU!**

Remember to

**Wash
Your
Hands**



SITE SAFETY - COVID-19	
All visitors and drivers must report to site office  Do not come on to site if you think you have coronavirus or someone in your household is self-isolating  No access for unauthorised personnel  Maintain social distances  Wash hands regularly	Children must not play on this site  Minimise face to face contact  Disinfect <u>ALL</u> vehicle surfaces regularly  Keep vehicle windows open for ventilation  Wear the correct PPE at all times
EMERGENCY PROCEDURES Please refer to our Health and Safety Policies	FIRE – If minor, immediately sound alarm and alert all those present within the premises and tackle same with on-site fire extinguishers. Call fire service if not brought under immediate control. If major, evacuate building and call Provincial Fire Service. Report ALL accidents to site safety officer and Project Manager. Notify Safety officer to investigate. ACCIDENT – Report immediately to Site Engineer and Project Manager. Action to be taken as per Haom Builders Health and Safety Policy.

17. APPENDIX NO: 03 PROJECT SITE LOGBOOK

HAOM BUILDERS

P. O. Box 797
Lae, 411
MP,
Papua New Guinea

Ph: (+675) 472 0625
Fax: (+675) 472 0601
Mb: 7033 6536
E-mail: haombuilders@gmail.com

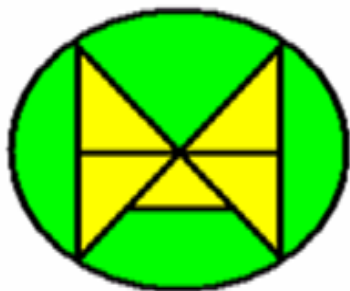
SITE DIARY

PROJECT:	CONSTRUCTION OF AGEVAIRU HEALTH CENTRES STAGE I & II					
CLIENT:	CENTRAL PROVINCIAL HEALTH AUTHORITY					
CONTRACT TYPE:	MAJOR WORKS					
MAIN-CONTRACTOR:	HAOM BUILDERS					
CONTRACT VALUE (PGK):						
DATE:						
SITE ENGINEER:						
MANPOWER				MACHINERY		
DESCRIPTION	MAN/ DAY	HRS	COST CODE	DESCRIPTION	MACHINE PER DAY	HOURL
Site Manager				Tipper Truck		
Supervisor				Backhoe		
Carpentry Foreman				Front End Loader		
Leading Hand Carpenters				Bulldozer		
Carpenter				Excavator		
Plumbing Foreman				Drum Roller		
Plumber				Generator		
Joiner Foreman				Plate Compactor		
Joiner				Vibrator		
Aluminum Fixer				Tampering rammer		
Boilermaker				Water Pump		
Welder						
Rigger				WEATHER CHART		
Steel Fixer				FAIR	1	
Gyprock Plasterer				CLOUDY	2	
Concrete Finisher				MODERAT E RAIN	3	
Tiler				HEAVY RAIN	4	
Painter				STORMY	5	
Electrical Sub-Contractor						
Storeman						
Site Security						
Laborers						

SUMMARY OF WORKS UNDERTAKEN:

CONTRACTORS ENVIRONMENTAL MANAGEMENT PLAN (CEMP)

[illegible]

18. APPENDIX NO: 04 COVID 19 MITIGATION AND MANAGEMENT POLICY**HAOM BUILDERS**

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Papua New Guinea

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Mb: 7033 6536
E-mail: haombuilders@gmail.com

DEPARTMENT OF HEALTH
PAPUA NEW GUINEA

**COVID 19 RISK ASSESSMENT & MANAGEMENT
PLAN**

PROJECT:	Construction and Completion of Agavairu Health Centre Stages I and II in Kairuku Hiri District, Central Province.			
EMPLOYER:	National Department of Health			
PROJECT:	Health Services Sector Development Project (HSSDP)			
COUNTRY:	Papua New Guinea			
PREPARED BY:	Jeremiah Rockey – Estimator/Senior Projects Engineer			
DUE DATE:	12 th of August, 2022			
HAOM BUILDERS		HSSDP		
PREPARED BY:	Mr. Jeremiah Rockey Senior Projects Estimator/Senior Projects Engineer	CHECKED BY:		DATE:
SIGHTED BY:	Mr. Hamo Anep (Projects Manager)	APPROVED BY:		DATE:
ISSUE 1:	For Client Approval			

18.1 INTRODUCTION

Since its emergence in late 2019, Covid 19 has resulted in many deaths across the globe, region and country. This disease has also halted and crippled supply and demand chain thus resulting in decline in the global economy for the whole of year 2020. Variants of the Covid 19 virus have been evolving in different region and has been spreading rapidly by movement of people around the world.

Papua New Guinea has recorded cases of Covid 19, Delta Variant and Omicron Strain of the virus. This virus inflicted diseases have also caused significant number of infections and deaths in most of the provinces in Papua New Guinea. Due to its mode of airborne transmission, it has spread rapidly in the country by movement of people. Lock downs has caused most businesses including construction contractors to stop work, thus delaying production, services and projects. This has resulted in most local contractors closing down as there was minimal economic activity to sustain the construction industry.

Some contractors including Haom Builders struggled during the pandemic to deliver milestone activities and the project on time due to supply and logistics constraints. The effect of this global pandemic has caused a number of challenges on our CHP Projects which included, delays, disruptions, uncertainty and increased costs.

This document will establish the approached which will be taken to manage and mitigate the risks of transmission of Covid 19 within Haom Builders construction sites and the surrounding communities. The Covid 19 Risk Assessment and Management Plan will form part of Haom Builders Contractors Environmental Management Plan Construction and Completion of Agavairu Health Centre Stages I and II in Kairuku Hiri District, Central Province.

The purpose of this Covid 19 Risk Assessment and Management Plan is to be used on Haom Builders construction sites as a guide to identify, contain, mitigate and minimize the spread of Covid 19 virus and its variants on site and surrounding communities. This document will cover the following topics:

- Risk Assessment and Management Plan
- Planning
- Site Operation Procedures
- Social Distancing
- Health Screening and Cleaning
- Quarantine & Access Restrictions
- Employees Wellbeing
- Community Interactions
- Supporting Project Affected Communities
- Anticipating Site Shutdowns
- Closing Sites

18.2 RISK ASSESSMENT & MANAGEMENT PLAN

Risk Description	L	C	R	Mitigating Measures	Responsibility
New strains or variants of Covid 19 will emerge in country	5	5	10	- Cease movement in and out of site; - Allow certain individuals to move in and out of site for urgent needs. - Regular temperature checks and decontamination before entry to site; - Impose 14 days isolation period on site for anyone travelling from outside of province where they are required to be isolated; - Make awareness and educate workers on new strains within site. - Any change in local and national government Covid 19 protocols and guidelines will be followed.	Site Specific Covid 19 Response Team, Workers
Covid 19 may spread among workers on site	5	5	10	- Site to be put on total lockdown with no entry or exits; - All workers will be isolated and Central PHA and Covid Response Team will be called evacuate patients to provincial isolation centres to carry out tests and treatments on individuals; - Every worker will go through temperature checks prior to enter common areas and crowded work areas; - Disinfect common utilities regularly on site; - Anyone showing signs and symptoms of infection will be immediately put to isolation; - Social distancing, compulsory mask wearing rules will be enforced.	Site Specific Covid 19 Response Team, Workers, Central PHA, Provincial Covid Response Team
Suspected case may be present on site	5	5	10	If a worker develops suspected COVID-19 symptoms the following should occur: - Worker to be put into isolation immediately on site. - Provincial Covid Response Team will be called into diagnose disease. - They must then follow the guidance on isolation and not return to use common facilities until their period of self-isolation has been completed. - The work area should receive deep cleaning and social distancing maintained. - A single point of contact is to be nominated for liaising with the authorities	Site Specific Covid 19 Response Team, Workers Central PHA, Provincial Covid Response Team
Travelling within Province and Country may get workers infected	5	5	10	- Worker will not travel if they are experiencing covid symptoms, received a positive covid test or have been informed by Provincial Covid Response Team; - Workers will be tested and cleared for travelling by established testing sites within province of origin. - When using public transport, it is strongly advised to wear face covering - Where corporate vehicles are shared (such as minibuses), limit passengers, increase ventilation and cleaning and encourage bubbles amongst staff (cohorting)	Workers
Uncontrolled Access/egress to	5	5	10	- An established and controlled point of entry and exit will be identified.	Site Specific Covid 19

CONTRACTORS ENVIRONMENTAL MANAGEMENT PLAN (CEMP)

site may introduce infections on site				- Log all visitors to the site and continue to utilize the temperature checks and hand sanitation stations. - Encourage all workers to maintain high levels of hand washing hygiene before entering or leaving the site - Regularly clean common contact surfaces in reception, office, access control and delivery areas. - Increase ventilation across site where possible - Continue to wear PPE in accordance with both site rules and your Health and Safety Policy	Response Team, Workers
Poor Personal Hygiene may lead to infections	5	5	10	- Wash your hands thoroughly and regularly. Use soap and water for at least 20 seconds. Use alcohol-based hand sanitiser or soap and water. - Avoid touching your face/eyes/nose/mouth with unwashed hands and cover your cough or sneeze with a tissue then throw it in the bin. - Regularly clean the hand washing facilities and check soap and sanitiser levels. - Provide suitable and sufficient rubbish bins for hand towels with regular removal and disposal. - Monitor levels of stock and top up accordingly. - Continue to enhance the cleaning regimes for toilet facilities, particularly door handles, locks and the toilet flush	Site Specific Covid 19 Response Teams Workers
Use of welfare facilities on site - exposure from large numbers of persons can make virus spread easily	5	5	10	- Dedicated eating areas should be identified on site to reduce food waste and contamination^[1] - Break times are encouraged to be staggered to reduce congestion - Hand cleaning facilities or hand sanitizer should be available - All areas used to receive regular cleaning - Increase ventilation in common areas where possible	Site Specific Covid 19 Response Teams
Manual handling - dual lifting (risk of transmission)	3	4	7	- Always consider if the task can be performed with one person using mechanical aids; - Ensure the individual(s) are fit for work prior to commencing task, showing no covid symptoms; - Break down the load where possible so that one person can comfortably carry; - Where teams are used, cohorting fixed teams is encouraged	Workers
First aid - including mental health	5	5	10	- First aid contents to be monitored to ensure adequate supplies remain - First aid cover arrangements to be reviewed - First aiders to ensure they have received refresher training so that any amended practices to keep people safe from covid is followed - Emergency plans on site and communicated so all staff understand what action to take in the event of a confirmed case of covid - Mental health first aiders (MFHA) to be implemented as required - Communicate any occupational health service available to the workforce including any available employee assistance programme (EAP) - Line management to regularly communicate to their team(s) - Effective reporting system established on site in order to rectify any raised issues or incidents in a timely manner	Site Specific Covid 19 Response Teams

Key

L = Likelihood (5= Almost certain, 4= Likely, 3= Possible, 2= Unlikely, 1= Rare)

C = Consequence (5= Severe, 4= Major, 3= Moderate, 2 = Minor, 1= Negligible)

R = Risk level (L + C)

18.3 PLANNING

18.3.1 Recruitment & Repatriation

One of the recruitment requirements will be to provide Covid 19 Vaccination Card or Covid 19 Tests Record/Clearance. Selected workers will be told to self-isolate at home for a week prior to repatriation. Covid 19 Tests will be carried out at the Covid Testing Sites in Lae and clearance obtained from Morobe Provincial Health Authority (MPHA) or Central Provincial Health Authority (CPHA) before repatriation of any group of workers to site.

18.3.2 Site Specific Covid Response Team

This team will prioritise implementation of Covid 19 Protocols on site. The team will consist of;

1. Local Community Health Worker at a nearest Health Centre – Check and diagnose workers with symptoms, make awareness within site and surrounding community;
2. Site Manager/Engineer – Administer and manage Covid 19 Risk Assessment and Management on site;
3. Site Supervisors/Leading Hands – Monitor implementation of Covid 19 Management Plan on site.
4. Site Safety and Environmental Safeguard Officer – Implement and enforce OHSE and Covid 19 Management Plan on site;
5. Site Nurse/First Aider – Administer treatments or first aids to workers injuries
6. Site Securities – Monitor movement in and out of Site, assist Site Safety and Environmental Safeguards Officer in implementation of daily temperature checks, hand washing and wearing of face masks;
7. Drivers – Standby for any emergency and evacuation in the event of presence positive cases on site.

18.3.3 Covid 19 Facilities on Site

Considerations will be given for Covid 19 facilities during site establishment. The following facilities will be made available although the construction period;

- Temperature Checks and Hand Sanitation Stations at Site Entry and Exit Points and common use facilities such as kitchen, ablution blocks, common room and amenities;
- 3x Bedroom Isolation Units for suspected cases or new arrivals on site. This facility will be barricaded and set aside for isolation purposes from other facilities;
- Daily surface cleaning and sanitizing of commonly used utilities such as benches, rails, door handles etc.
- Emergency muster points and evacuation points on site;
- Establish effective contact with Provincial Covid 19 Response Team for testing or evacuation of confirmed cases.

18.3.4 Re-evaluate Risk Frequencies

Risks will be frequently evaluated as circumstances change. The Site Specific Covid 19 Response Team will be tasked to review, update and generate safe working procedures, prepare for emergency and response plan to address risks. They will also develop procedures, recovery plans and contingency measures ready if confirmed cases occurs on site.

18.3.5 Continuity of Work on Site

Minimum number of labour and supervision will be kept to carry out critical activities in the event of spread of Covid 19 on site or prolonged nationwide lockdowns. Works programme will be revised to focus on critical construction activities as well as critical OSHE and Covid 19 Management activities.

18.3.6 Site Operation Procedures

18.3.6.1 Mandatory Temperature Checks on Site

Temperature checks will be made mandatory on all entry points into the site, common use facilities. Workers will be strictly monitored.

18.3.6.2 Face Masks

Face masks wearing will be mandatory in all crowded area including vehicles. Exception will be given for removal of face masks during meal times. Workers will be strictly monitored.

18.3.6.3 Hand Sanitizers

Hand sanitizers or water and soap will be placed at all sanitizing stations on site. Workers will also be encouraged to have their own sanitizer with them at all times.

18.3.6.4 Social Distancing

Workers will be told to practice and maintain social distancing. 2m of distance will be maintained. Close contact works, for working in confined spaces evaluated to minimize overcrowding. Workers will be instructed to wear masks and cover their face when coughing.

Virtual meetings will be attended by restricted number of workers to cater for social distancing space. Meetings can be held one at a time. All site activities will be planned to cater for social distancing.

Camp population densities will be minimized as much as possible to facilitate social distancing. For example, number of people in workers accommodation will be reduced to a worker per room. Room partitions will be created for each worker.

Workers will be informed to practise social distancing during toolbox meetings, trainings and awareness. Workers will be discouraged to attend large gatherings outside project site or when in towns.

18.3.6.5 Quarantine and Access Restrictions

A quarantine or containment facility will be established on site. If a worker within a dormitory is infected, the whole dormitory will be put on quarantine. Local Covid Response Team will be called in to monitor and control those who are under quarantine.

Access on site will be restricted to site personals, deliveries, site inspections and stakeholders' meetings. Entry to site will follow site protocols. Outsiders and locals will be highly restricted to enter site or visit workers accommodations at any one time. All entrance will be contact less. All gates will be opened and closed by guards. All occupants of delivery vehicles will be told to remain in their vehicle unless otherwise needed to unload. Social distancing will take effect if the later occurs.

18.3.6.6 Communication

Good communication within site and surrounding community is essential to effectively identify hazards and control risks. It is vitally important all workers, starting from Project Manager, Site Manager, Site Supervisors and Leading Hands engage and consult with workers.

Mediums such as posters, videos and signs will be put up on site for all to see and be educated. Workforce on site will be constantly reminded about the protocols in place on site during inductions, toolbox and safety meetings. Alternative means of communication would be social media, radio, PA systems mounted on vehicles and leaflets or posters.

18.3.6.7 Employee Wellbeing

There will be regular transparent communications to reassure and update workforce in the event of lockdowns. Additional options at workers camps like televisions or issuing of Flex Cards or communication lines will be open for workers to access. Bonuses for working during lockdowns will also be considered. Any entertainment including alcohol will be strictly banned on site. Relevant counselling groups will be organized to support emotional wellbeing of workers.

18.3.6.8 Supporting of Project Affected Communities

Goal is to minimize risk of spreading virus within project impact communities and the construction site. Local communities' initiatives will be support in the event of re-emergence of Covid 19. Water. This may include provision of soap, water and hand sanitizers will be made available at public places. Site vehicle will be made available for medical evacuation or logistics for communal purposes.

18.3.6.9 Anticipating Site Shutdown

In the event that national or international lockdowns are prolonged and economic activities come to standstill and Haom Builders could not sustain construction activities on site, shut down will be imposed. Shut down will be done upon clients or project owners' approval after all contractual obligations are met under this contract.

Operations on site will be scaled down to completion of critical tasks using materials at hand. Work will be done in a safe and responsible manner to minimize risks of infection to remainder of workers. No new activities will be started. Current activities safely brought to a stop. Preservation of completed works will be done. All these shut down activities will confirm to ethical probity contract and procurement rules set out in the contract.

18.3.6.10 Closing Site

If given the approval to close down site, general clean-up will be done to ensure site is left in a tidy condition. Haom Builders will ensure all remaining materials and works are protected to last for up to 6 months. Site amenities will be cleaned and in order and locked up. Site will be left in a condition in which cost and effort of remobilization is minimized. All mobile plants and equipment will be covered properly to avoid damage by natural weather condition or vandalism. Skeleton staff will remain on site to look after the site.

Arrangements will be made provide temporary layoff incentives or paid leave for those workers who will be laid off from work. Client will be notified on remaining payments and balances owed to contractors and suppliers so all debts are settled as soon as possible.